

Regional Financing Agreements and Currency Swaps: Impact on Investors' Confidence

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AEA Summer Program

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Research Question: How does signing a *Regional Financing Agreements(RFA)* and *Bilateral Currency Swaps* affect bond spread of a country?

What is Bond Spread?

➤ **Bond Spread** : the difference between the quoted rate of return of different government bonds.

Bond spread = Yield of the foreign government bond - Yield of US treasury bond

For example:

If you are an investor, would you invest your money in economically unstable countries with risk?

It is only yes when the return is big enough!!!

Source: <https://www.investopedia.com/terms/y/yieldspread.asp>

What are RFAs and Swaps?

➤ ***Regional Finance Agreements (RFAs)***

Definition: RFAs are mechanisms or agreements through which groups of countries mutually pledge financial support to countries experiencing financial difficulties in their regions.

Purpose: The primary purpose of RFAs is to enhance regional financial stability and to complement global financial safety nets provided by institutions like the International Monetary Fund (IMF).

Source: <https://www.esm.europa.eu/content/what-are-regional-financing-arrangements-rfas>

What are RFAs and Swaps?

Example:

European Stability Mechanism (ESM)

Chiang Mai Initiative Multilateralization (CMIM)

Latin American Reserve Fund (FLAR)

Arab Monetary Fund (AMF)

What are RFAs and Swaps?

➤ ***Bilateral Currency Swaps:***

A currency swap is a financial agreement between two parties to exchange specified amounts of different currencies and then reverse the exchange at a future date, often including periodic interest payments in the respective currencies during the life of the swap.

Example:

Nigeria-China Swap, 2018

Pakistan-China Swap, 2011

We want to see the impact of RFAs and bilateral swaps on investors confidence.

➤ **How can we see investors' confidence?**

- Spreads against treasury bonds are a measure of investors' confidence about a country's macroeconomic stability
- Narrowing spread means that investors are confident about the country's ability to pay their commitments to both domestic and foreign investors
- we are analyzing if signing a RFA/swaps would narrow the spread in certain time periods after it was signed

Source: <https://www.federalreserve.gov/econresdata/notes/feds-notes/2016/recession-risk-and-the-excess-bond-premium-20160408.html>

➤ Data Sources:

- [Global Financial Data](#)
 - Contains bond spreads of nearly all countries
- [Global Financial Safety Net](#)
 - Contains details of RFA agreements and Bilateral Swaps
- [World Bank Open Data](#)
 - Contains details about risk premium on certain countries debt

Methodology

➤ **Models of Interest:**

- Difference in Differences
 - Countries with similar economic positions(for example: similar GDP per capita, area and population)
- Regression Discontinuity
 - Time-invariant regression discontinuity with time as a running variable

➤ Variables of Interest:

- Xs: Membership in a RFA (dummy), other confounding variables
- Y: Bond Spread/Goldman Sachs Index/JP Morgan Chase Developing Country Index

$$BondSpread_{it} = \alpha + \beta RFA_{it} + \gamma POST_{it} + \lambda RFA * POST_{it} + \varepsilon_{it}$$

➤ Country pairs for Difference in Differences:

RFA Differences:

- Sweden and Norway
- Costa Rica and Panama, Honduras, El Salvador
- Guyana and Ecuador
- Azerbaijan and Armenia

Bilateral Swaps Differences:

- Vietnam and Malaysia, Thailand
- Kenya and Nigeria

Our Hypothesis:

Having a RFA or a Bilateral Swap should narrow the bond spread

Questions?

References

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Thank you!